

STAT 405 Final Project Report

Car Crashes, Weather, and Pedestrians in Vancouver

Ratana Wang (45508256), Victor Liu (80536576)

GitHub repo: <https://github.com/ratanawang/stat405>

1 Introduction

Pedestrians are the most at-risk group for traffic accidents. In B.C., over 2000 pedestrian injuries occur on average per year, with the vast majority (79%) of these accidents occurring at intersections [ICBC, b]. Pedestrian fatalities are likewise not uncommon; the City of Vancouver found that pedestrians made up over 60% of traffic accident fatalities in 2021, where nearly half of those fatalities involved senior pedestrians aged 65 and above [City of Vancouver]. Weather conditions—which fluctuate by season and are especially volatile in Vancouver—and time of day also exacerbate the car crash risk. There is room for improvement in understanding and mitigating this risk, and so in this report, we investigate using Bayesian methods the **probability of pedestrian involvement in car crashes, given the spatial (road intersection), temporal (time of day/day of week), and environmental (weather) contexts.**

2 Literature Review

Our review of the existing literature finds the following key points. First, pedestrian safety at intersections remains a worsening problem worldwide; however, since this is a complex issue where numerous factors are in play, region-specific and even intersection-specific trends should be studied to provide the most effective reactive fixes [Haque et al., 2025].

Next, Bayesian frameworks are popular in car crash analyses, but there is a gap in the study of the *combined* effect of spatiotemporal and weather data on pedestrian involvement specifically. For instance, Song et al. [2020] use a hierarchical random-effects model to study pedestrian injury severity in car crashes in North Carolina, with a focus on exploring only spatiotemporal effects. Cheng et al. [2017] similarly use a Bayesian multivariate crash frequency model, but focus on motorcycle—rather than pedestrian—incidents and weather effects. In Poland, Filapek et al. [2025] recently found through Bayesian Poisson regression that temperature and humidity were helpful in explaining short-term fluctuations in general crash frequency and that, interestingly, socioeconomic factors (e.g. car registrations) played a much stronger role in predicting accident rates over the long term.

To our surprise, Filapek et al. [2025] and Cheng et al. [2017] both found precipitation (rainfall) to have a minimal effect in explaining traffic accidents, so we are curious to see how our results may differ in a rainy city like Vancouver.

The datasets we used, which are local to Vancouver, are publicly available. Existing car crash location analyses report simple crash counts based on location (e.g. Adibi [2020] and Vision Zero Vancouver), but do not take into account other conditions (weather, time, etc.) nor have a particular pedestrian focus.

With a need for region-specific research and further understanding of not only spatial/temporal but also environmental effects on car crashes involving pedestrians, we hope to surface results that will help Vancouver become a more walkable city.

3 Data Preprocessing

In this report, we use two main sources of data: (1) ICBC car crash data, available via Kaggle and an official Tableau source for ICBC [a]; and (2) Government of Canada weather data measured from the Vancouver International Airport. These data span the years 2018 through 2024, which helps us gather a more complete picture of both pre- and post-COVID-19 pandemic trends.

We clean, preprocess, and combine these data using a custom R script `data.R`. For the ICBC data, we remove overlapping years, rename columns to facilitate joining, and filter for the municipality of Vancouver only. Similarly for the weather data, we merge data across all years and transform columns into aggregate features (e.g. `avg_rain_mm`). We also select for only the relevant variables before combining. The result of this script is a `crash_weather.csv` file that we use as a basis for the rest of our report.

We centered and standardized the latitude and longitude variables, given that their magnitudes are large but very close to each other, which helped our MCMC methods run faster. We also centered the Intersection indicator variable, because we found that it helped our custom MH sampler for Model 1 mix better. See Appendix C for details.

4 Bayesian Models

To answer our research question, we consider two different Bayesian models. The first one is simple, focusing only on intersection and weekend effects on pedestrian involvement, while the second one is more complex and takes into account more factors (e.g. weather).

We use the following notation across our two models:

$Y_i \in \{0, 1\}$	1 if crash i involved a pedestrian, 0 otherwise
$I_i \in \{0, 1\}$	1 if crash i occurred at an intersection, 0 otherwise
$W_i \in \{0, 1\}$	1 if crash i occurred on a weekend, 0 otherwise
$\text{logit}^{-1}(u) = (1 + e^{-u})^{-1}$	the <code>inv_logit</code> function

4.1 Model 1: Simple Baseline with Few Parameters

For our first model, we consider a very simple subset of features—whether the crash happened at an intersection, and whether the crash happened on a weekend (Saturday or Sunday). We also have the intercept parameter to estimate the baseline log-odds of a pedestrian being involved in a crash. The model is defined using mathematical notation below¹:

$$\begin{aligned} \text{Priors: } & \beta_0 \sim \mathcal{N}(0, 5), \quad \beta_I \sim \mathcal{N}(0, 2), \quad \beta_W \sim \mathcal{N}(0, 1) \\ \text{Likelihood: } & Y_i \sim \text{Bern}(\text{logit}^{-1}(\beta_0 + \beta_I I_i + \beta_W W_i)) \end{aligned}$$

4.1.1 Prior Choice Motivation

All parameters are real-valued because they may have positive/negative effects on pedestrian involvement. We centered all parameters ($\beta_0, \beta_I, \beta_W$) at 0, since we have no prior information to believe strongly in either direction, and zero would indicate the parameter has no effect on the log-odds of pedestrian involvement in a crash. From our EDA, we found that only around 2% of reported car accidents involve pedestrians, for which the corresponding log-odds value is about -4, so we chose a larger standard deviation of 5 for the prior of baseline parameter β_0 so that it can move more easily toward that value without strictly enforcing it.

¹We use the $\mathcal{N}(\mu, \sigma)$ version, with standard deviation instead of variance, when specifying Normal distributions.

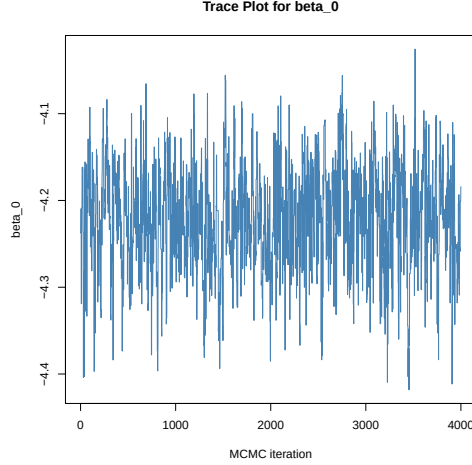


Figure 1: Trace plot for β_0 shows “hairy caterpillar”, indicating a successfully mixed chain.

4.1.2 Posterior Approximation Methods

For Model 1, we used two different methods to approximate the posterior: (1) a custom Metropolis-Hastings (MH) sampler, and (2) variational inference (VI) in Stan.

Our custom MH sampler uses alternating kernels. First, kernel K_1 proposes a new value for β_0 , and it is accepted or rejected according to the MH ratio as usual. Then, kernel K_2 proposes a new value for β_I , following the same process. Finally, kernel K_3 proposes a new value for β_W . As discussed in class, this algorithm remains π -invariant. Our custom MH sampler runs for 5000 iterations, 1000 of which are used for the burn-in period, leaving 4000 samples per run.

As we were testing our custom MH sampler, we generated trace plots to see if the chain was mixing well. In our first few runs of the algorithm, the trace plots did not show good mixing, which led us to fine-tune the proposal standard deviations. After many tunings, we found a set of proposal SDs that showed fast mixing and good ESS. The histograms for the posterior distributions of the parameters also appeared to be approximately normal. See details in Appendix B.

The second method we used to approximate the posterior for Model 1 was VI in Stan. See Appendix C for the code. One key finding was that VI is significantly faster than our custom MH sampler: while the custom MH sampler took around 1-2 minutes to complete, VI only took around 10 seconds.

When comparing the posterior distributions from the custom MH sampler and from VI, we found that the summary statistics (such as mean and standard deviation) for each parameter were similar between the two methods. This gives us more assurance that both methods provide an adequate approximation to the posterior distributions of all the parameters. Because VI is much faster, we find that, of the two methods, VI is better overall for Model 1.

4.1.3 Correctness Check of Custom MH

To test the correctness of our custom MH sampler, we used the Exact Invariance Test, which involves performing many iterations of MCMC initialized from synthetic data generated from forward simulation. The code for this can be found in Appendix C, or in the GitHub repo. For both the two-sample t -test and two-sample Kolmogorov-Smirnov test, we found that the p -values were well above 0.1, indicating that there is no evidence of a bug in our custom MH sampler code.

4.2 Model 2: Bernoulli Regression with More Parameters

For our second model, we consider a larger and richer subset of features, which include all features from Model 1 (baseline, weekend, intersection), as well as the longitude/latitude of crash location, average amount of rain (in mm) and snow (in cm), and time of day. Although this model uses many parameters, we use a large subsample size ($N \approx 29000$; 10% of the full data) that reduces the risk of overfitting.

The model is defined using mathematical notation below:

$T_{ij} \in \{0, 1\}$	1 if crash i occurred in time category j , 0 otherwise
$t_i \in \{0, 1\}^K$	length K vector of T_{ij} values for crash i
$\text{snow}_i \in \mathbb{R}^+$	monthly average snow amount, in cm, for crash i
$\text{rain}_i \in \mathbb{R}^+$	monthly average rain amount, in mm, for crash i
$\text{lon}_i \in \mathbb{R}$	standardized longitude of crash i site
$\text{lat}_i \in \mathbb{R}$	standardized latitude of crash i site

The K time categories combine existing time buckets, and are as follows²:

- $j = 0$: afternoon rush, 3:00 PM - 6:00 PM
- $j = 1$: evening, 6:00 PM - 12:00 AM
- $j = 2$: late night, 12:00 AM - 6:00 AM
- $j = 3$: midday, 9:00 AM - 3:00 PM
- $j = 4$: morning rush, 6:00 AM - 9:00 AM

Priors: $\beta_0 \sim \mathcal{N}(0, 5)$, $\beta_I \sim \mathcal{N}(0, 2)$, $\beta_W \sim \mathcal{N}(0, 1)$, $\beta_t \sim \mathcal{N}_K(0, 1)$,
 $\beta_{\text{snow}} \sim \mathcal{N}(0, 3)$, $\beta_{\text{rain}} \sim \mathcal{N}(0, 3)$, $\beta_{\text{lon}} \sim \mathcal{N}(0, 2)$, $\beta_{\text{lat}} \sim \mathcal{N}(0, 2)$

Likelihood: $Y_i \sim \text{Bern}(\text{logit}^{-1}(\beta_0 + \beta_I I_i + \beta_W W_i + t_i^\top \beta_t + \beta_{\text{snow}} \text{snow}_i + \beta_{\text{rain}} \text{rain}_i + \beta_{\text{lon}} \text{lon}_i + \beta_{\text{lat}} \text{lat}_i))$

Note that some of these priors are justified by the same reasons as for Model 1. For new parameters, we chose arbitrary small standard deviations. All prior distributions are Normal because, again, each parameter may positively or negatively affect pedestrian involvement in car crashes.

4.2.1 Posterior Approximation Methods

For Model 2, we likewise use two different methods to approximate the posterior: (1) Hamiltonian Monte Carlo (HMC) in Stan, and (2) variational inference (VI) in Stan.

First, we ran 2000 post-burn-in iterations using Stan, which uses the NUTS algorithm (based on HMC). To evaluate whether HMC in Stan was working properly, we looked at several diagnostics, including trace plots for all parameters; they all show a “hairy caterpillar” that indicates the chain is mixing well. See Appendix B for details. We also looked at the effective sample size (ESS) which, for all parameters, ranged from around 900 to 3000—rather large when compared with our 2000 post-burn-in iterations. This also indicates good mixing.

For the second method, we used VI in Stan. Similar to Model 1, we found that VI was significantly faster compared to the other method. HMC took around 3-4 minutes to run, whereas VI took under 10 seconds. To evaluate whether VI approximated the posterior well, we compared its summary statistics (mean, median, sd, etc) to that of our HMC run, which were quite similar for all parameters of our model. This indicates

²These time categories are in alphabetical order.

that VI provides a good approximation to the posterior for a fraction of the computational cost, leading us to conclude again that VI is the better method to use here. See Appendix A for these statistics.

4.3 Results and Interpretation

In both of our models, each parameter tells us how much a given factor (intersection, weekend, rain, snow, etc.) affects the probability of pedestrian involvement in a crash. However, the effects of each factor are not equal; our models found that some factors were more relevant than others.

Table 1: Model 1 variational inference summary (`model1_vi_summary.csv`).

variable	mean	median	sd	mad	q5	q95
<code>lp_</code>	-2555.726	-2554.701	3.172	2.032	-2562.062	-2552.879
<code>lp_approx_</code>	-1.477	-1.166	1.198	0.992	-3.862	-0.177
<code>beta_0</code>	-4.175	-4.174	0.050	0.048	-4.258	-4.092
<code>beta_i</code>	1.481	1.481	0.128	0.126	1.269	1.692
<code>beta_w</code>	0.163	0.161	0.093	0.094	0.011	0.316

The intercept parameter β_0 represents the baseline log-odds of pedestrian involvement, when we set all other variables to zero. Table 1 (above) shows that, under Model 1 VI, β_0 has a posterior mean of -4.17 , and Table 3 (in Appendix A) shows that, under Model 2 VI, the posterior mean is -4.67 . When converted to probabilities, one can interpret these by saying that the baseline probability of pedestrian involvement in car crashes (assuming all other variables are set to zero) is 1.5% and 0.92% under the two models respectively.

In both models, the Intersection indicator (whether or not the collision occurred at an intersection) increased the log-odds of pedestrian involvement the most, with the posterior mean of β_I estimated to be 1.48 under Model 1 VI and 1.47 under Model 2 VI. This means that the log-odds of pedestrian involvement is 1.48 higher for a collision occurring at an intersection compared to one that is not. Alternatively, it also corresponds to an odds ratio of $\exp(1.48) \approx 4.39$, meaning that under Model 1, the odds of a pedestrian being involved in a crash are approximately 4.39 times higher at an intersection than at a non-intersection. All other factors showed a weaker effect on the log-odds of pedestrian involvement.

For our four time-of-day parameters in Model 2, we found that their posterior means were all positive (with the exception of β_{t_3} , corresponding to midday, having a posterior mean near 0), indicating that the “baseline” time category with no parameter (Afternoon Rush) and the Midday time category had lower log-odds of pedestrian involvement. β_{t_2} , corresponding to Late Night, had the highest posterior mean at 0.89, indicating that late-night is the time period associated with the greatest increase in log-odds of pedestrian involvement. Meanwhile, the remaining time-of-day parameters β_{t_1} and β_{t_4} , corresponding to Evening and Morning Rush respectively, had posterior means of 0.49 and 0.38, indicating that these two time periods are associated with a smaller increase in log-odds compared to the baseline of afternoon rush hour.

The weekend parameter β_W had a mean of 0.16 under Model 1 VI and 0.11 under Model 2 VI, indicating little effect on the log-odds. Additionally, under Model 2, the 90% credible interval for β_W contains zero, which indicates that zero weekend effect is plausible.

The latitude and longitude parameters β_{lat} and β_{lon} had posterior means of 0.18 and -0.07 respectively under Model 2 VI, indicating that the North side of the city has slightly higher log-odds of pedestrian involvement, and similar (but less so) for the West side of the city.

Surprisingly, we found that weather factors (rain and snow) showed a weaker change in the log-odds than we earlier hypothesized. For example, under Model 2 VI, the posterior mean of β_{rain} is estimated to be 0.115. With the rainiest month in our dataset having a daily average of 10.41 mm of rain, this means that the change in log-odds for that month would be $0.115 \times 10.41 = 1.197$ (when compared to a month that has no rain), which is still less than the effect of being at an intersection. Meanwhile, the posterior mean of β_{snow}

is estimated to be -0.12 under Model 2 VI, which can be interpreted as meaning that an extra centimetre of snow decreases the log-odds of pedestrian involvement by 0.12. This is perhaps our most surprising result in this report, and echoes the findings of Filapek et al. [2025] and Cheng et al. [2017] in other contexts.

The full summary statistics for each parameter’s approximate posterior under each model can be found in Appendix A.

5 Discussion

5.1 Limitations

Unfortunately, we did not have greater precision weather data available for each crash, since ICBC data do not specify accident dates—they only specify the month in which they occurred. Thus, we were restricted to using only the monthly averages for weather data (e.g. for a car crash in March 2024, we use the average weather data across the entire month).

Another key limitation is how we fit all models on a reproducible simple random 10% sample of our crash data for the sake of computational tractability (i.e. using the full dataset for MCMC took way too long). This choice introduced more uncertainty to our posterior approximation, and since we did not do stratified sampling, the 10% sample may not be the most accurate representation of the full data. See Appendix C for details.

5.2 Implications

Pedestrians in Vancouver may find it useful to know what kinds of weather conditions, intersections, and timings put them more at risk for car crashes. Our project findings are also insightful for drivers, who benefit from knowing when and where to take more caution, as well as government civil engineers, who may target specific intersections for better design and infrastructure improvements.

5.3 Further Work

Future studies may want to look at the distribution of pedestrian injury severity in car crashes, instead of only predicting involvement; this was a weakness of our ICBC dataset, which did not record severity. However, other jurisdictions may have such data available.

Also, instead of using a simple “Intersection” indicator variable as we did in our two models, we could instead try a hierarchical model that estimates intercepts for individual intersections, e.g. Cambie & Broadway, and uses partial pooling. However, since there are numerous intersections in the city of Vancouver, this would require a large amount of hyperparameters, thereby increasing our computational cost. We would also need to ensure that the intersections are properly recorded and consolidated; the data may contain minor differences between intersection names (e.g. “Cambie St & W Broadway” vs. “W Broadway & Cambie St” vs. “Cambie St and W Broadway” vs. “Cambie Street and West Broadway”) that all refer to the same intersection. Right now, our latitude and longitude predictors play a role in the spatial analysis of crashes and pedestrians, but they assume that pedestrian risk is linear. This may be useful for capturing general east-west/north-south trends but may not be able to capture the full effects of central intersections.

6 Contributions

The two members of our team split the work fairly evenly and communicated frequently to ensure mutual understanding of the other’s work. We worked together in project ideation, model design, data preprocessing, and report writing. Victor handled more of the model implementation; Ratana handled more of the background research.

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Appendix A Tables

Table 2: Model 1 custom MH posterior summary (`model1_custom_mh_summary.csv`).

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
<code>beta_0</code>	-4.222	-4.221	0.059	0.059	-4.324	-4.126	1.004	392.567	531.657
<code>beta_i</code>	1.429	1.428	0.132	0.135	1.217	1.652	1.003	414.982	665.985
<code>beta_w</code>	0.145	0.147	0.100	0.104	-0.021	0.310	1.002	572.307	790.993

Table 3: Model 2 variational inference summary (`model2_vi_summary.csv`).

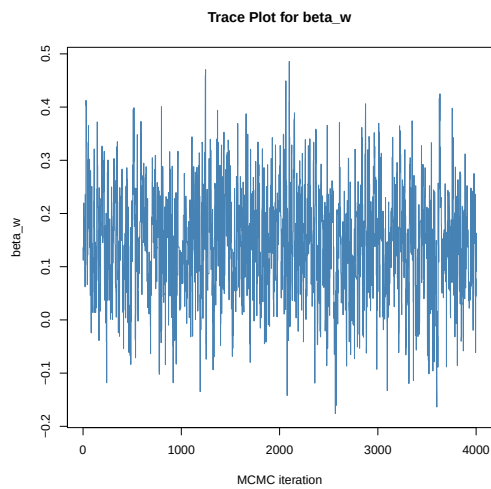
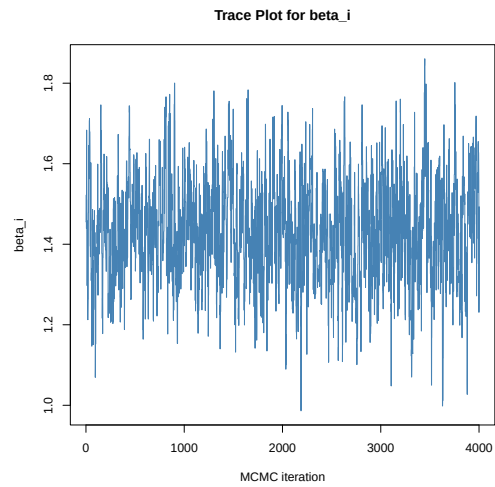
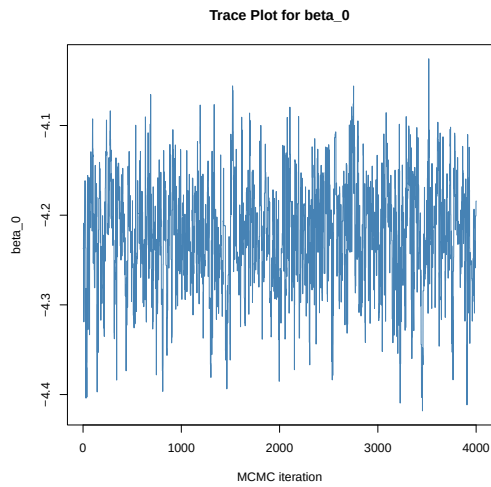
variable	mean	median	sd	mad	q5	q95
<code>lp_</code>	-2525.365	-2523.288	7.505	5.845	-2540.226	-2517.402
<code>lp_approx_</code>	-5.439	-5.076	2.279	2.153	-9.378	-2.424
<code>beta_0</code>	-4.675	-4.675	0.059	0.058	-4.771	-4.579
<code>beta_t[1]</code>	0.486	0.488	0.098	0.101	0.323	0.642
<code>beta_t[2]</code>	0.888	0.890	0.168	0.164	0.613	1.156
<code>beta_t[3]</code>	-0.007	-0.004	0.078	0.079	-0.135	0.121
<code>beta_t[4]</code>	0.381	0.383	0.117	0.115	0.190	0.570
<code>beta_i</code>	1.471	1.471	0.091	0.095	1.319	1.621
<code>beta_w</code>	0.109	0.111	0.080	0.080	-0.025	0.243
<code>beta_snow</code>	-0.123	-0.126	0.122	0.122	-0.328	0.075
<code>beta_rain</code>	0.115	0.115	0.009	0.010	0.100	0.131
<code>beta_lat</code>	0.182	0.183	0.048	0.048	0.102	0.259
<code>beta_lon</code>	-0.067	-0.067	0.040	0.041	-0.132	-0.001

Table 4: Model 2 HMC posterior summary (`model2_hmc_summary.csv`).

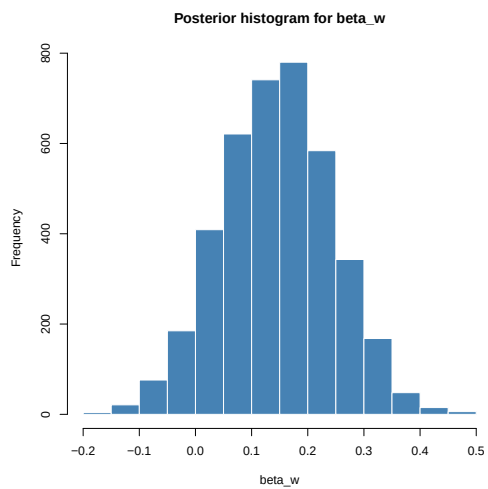
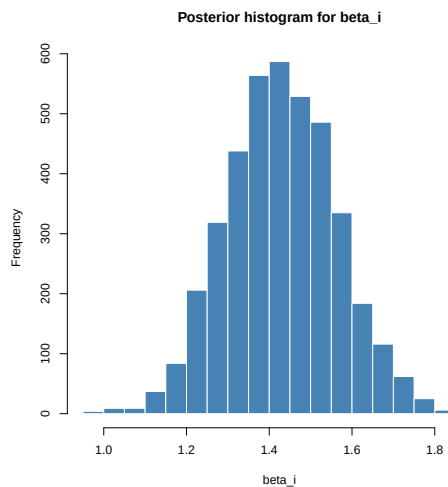
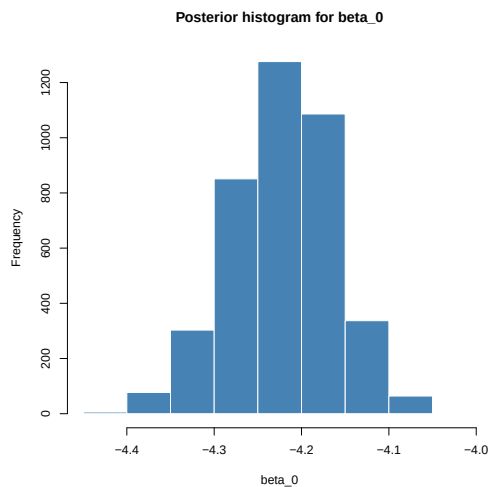
variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
<code>lp_</code>	-2503.978	-2503.619	2.355	2.214	-2508.525	-2500.764	1.000	914.661	1277.527
<code>beta_0</code>	-4.700	-4.700	0.123	0.122	-4.908	-4.504	1.001	1044.358	1349.046
<code>beta_t[1]</code>	0.461	0.460	0.127	0.123	0.248	0.674	1.004	1410.709	1537.723
<code>beta_t[2]</code>	0.906	0.910	0.203	0.206	0.579	1.228	1.001	1885.480	1521.871
<code>beta_t[3]</code>	-0.059	-0.060	0.119	0.123	-0.258	0.128	1.000	1401.180	1688.059
<code>beta_t[4]</code>	0.404	0.405	0.160	0.165	0.139	0.664	1.001	1792.224	1507.435
<code>beta_i</code>	1.437	1.438	0.125	0.124	1.236	1.644	1.000	2552.275	1451.383
<code>beta_w</code>	0.111	0.112	0.106	0.106	-0.061	0.290	1.001	2895.093	1338.264
<code>beta_snow</code>	-0.202	-0.200	0.138	0.139	-0.432	0.017	1.001	2769.817	1398.248
<code>beta_rain</code>	0.096	0.097	0.018	0.018	0.067	0.125	1.000	2291.861	1519.973
<code>beta_lat</code>	0.211	0.209	0.048	0.047	0.135	0.291	1.000	2975.095	1391.064
<code>beta_lon</code>	-0.075	-0.075	0.047	0.047	-0.150	0.003	1.001	2589.601	1566.843

Appendix B Plots and Diagrams

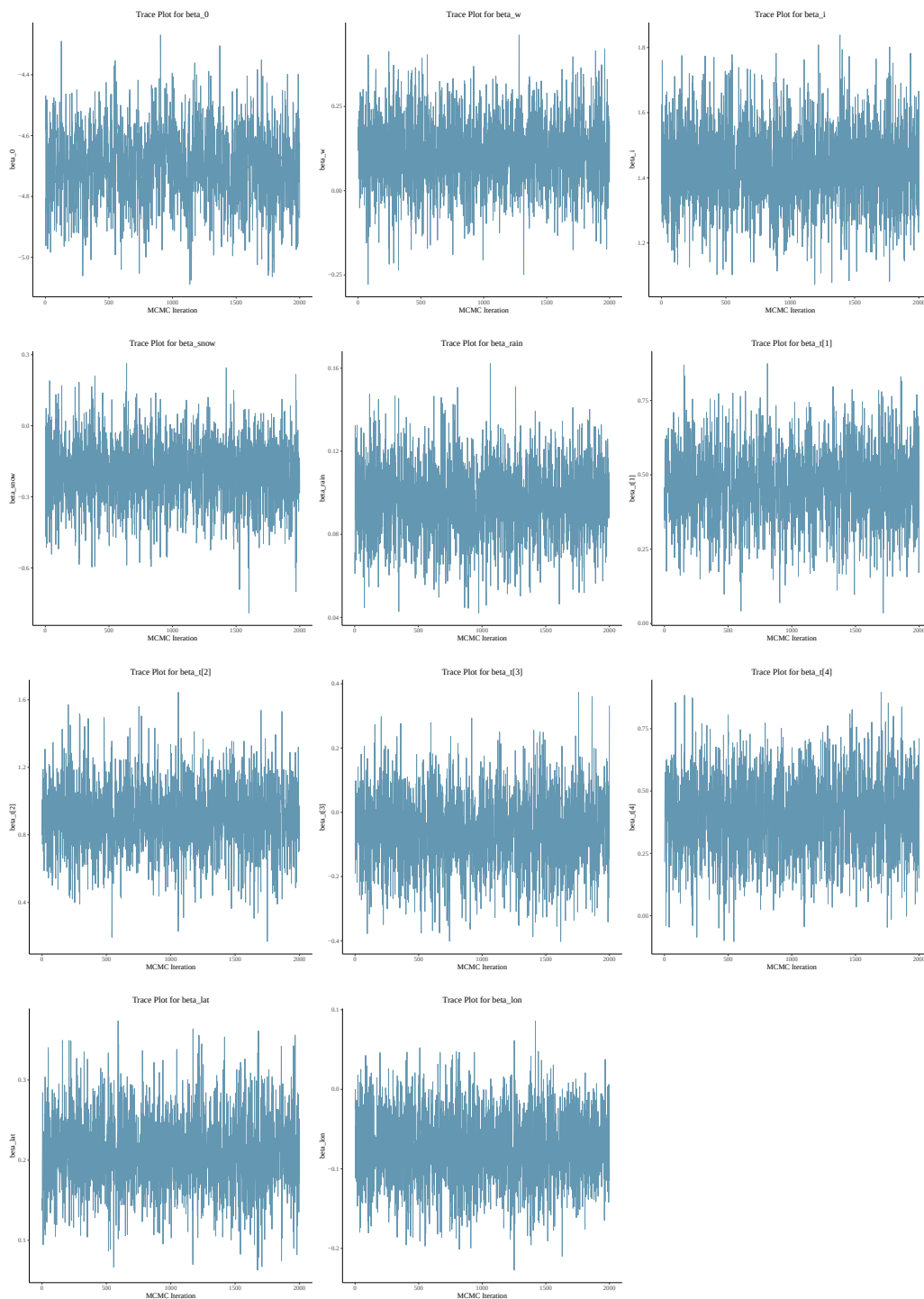
This page contains the trace plots for our custom MH sampler for each of Model 1's three parameters. The trace plots show that the chain mixes well.



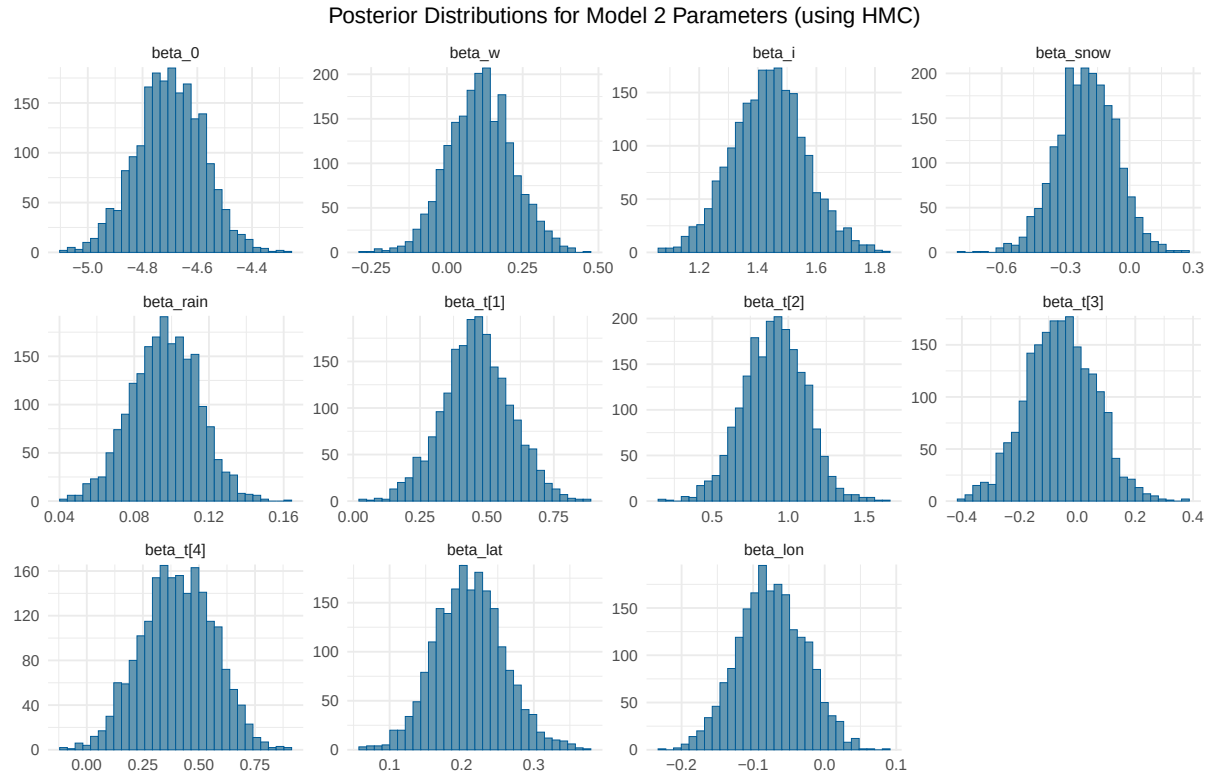
This page contains the histograms of the posterior distributions for our custom MH sampler for each of Model 1's three parameters.



This page contains the trace plots for HMC in Model 2, for each of Model 2's parameters. The trace plots all show that the chain mixes well.



This page contains the histograms for the posterior distributions for all parameters of Model 2 as estimated by HMC.



Appendix C R/Stan Code

Some portions of the code (e.g. print statements, loading packages) are omitted; the full code can be found in the GitHub repo. The data preprocessing file is not included here but it is still in the repo.

The first file here, `model1_custom_mh_functions.R`, provides the custom MH sampler as well as the functions need to perform the Exact Invariance Test.

```
1  # model1_custom_mh_functions.R
2
3  log_joint <- function(beta_0, beta_i, beta_w, y, intersection, weekend) {
4    # log prior
5    log_prior_beta_0 <- dnorm(beta_0, mean = 0, sd = 5, log = TRUE)
6    log_prior_beta_i <- dnorm(beta_i, mean = 0, sd = 2, log = TRUE)
7    log_prior_beta_w <- dnorm(beta_w, mean = 0, sd = 1, log = TRUE)
8
9    log_prior <- log_prior_beta_0 + log_prior_beta_i + log_prior_beta_w
10   log_odds <- beta_0 + (beta_i * intersection) + (beta_w * weekend)
11
12   # log likelihood
13   log_likelihood = sum(dbinom(y, size = 1, prob = plogis(log_odds), log = TRUE))
14
15   log_prior + log_likelihood
16 }
17
18
19
20 custom_mh <- function(y, intersection, weekend, n_iters, beta_0, beta_i, beta_w) {
21   start_time <- Sys.time()
22
23   beta_0_trace <- rep(NA, n_iters)
24   beta_i_trace <- rep(NA, n_iters)
25   beta_w_trace <- rep(NA, n_iters)
26
27   curr_beta_0 <- beta_0
28   curr_beta_i <- beta_i
29   curr_beta_w <- beta_w
30
31   curr_log_joint <- log_joint(curr_beta_0, curr_beta_i, curr_beta_w, y, intersection,
32     ↪ weekend)
33
34   acceptance_beta_0 <- 0
35   acceptance_beta_i <- 0
36   acceptance_beta_w <- 0
37
38   sd_beta_0 <- 0.1
39   sd_beta_i <- 0.15
40   sd_beta_w <- 0.15
41
42   for (i in 1:n_iters) {
43     # kernel 1: propose a new beta_0
44
45     beta_0_proposal <- curr_beta_0 + rnorm(n=1, mean=0, sd=sd_beta_0)
```

```

45
46 proposal_log_joint <- log_joint(beta_0_proposal, curr_beta_i, curr_beta_w, y,
  ↪ intersection, weekend)
47 log_acceptance_prob <- proposal_log_joint - curr_log_joint
48
49 if (log(runif(1)) < log_acceptance_prob) {
50   curr_beta_0 <- beta_0_proposal
51   curr_log_joint <- proposal_log_joint
52   acceptance_beta_0 <- acceptance_beta_0 + 1
53 }
54
55 beta_0_trace[i] <- curr_beta_0
56
57 # kernel 2: propose a new beta_i
58
59 beta_i_proposal <- curr_beta_i + rnorm(n=1, mean=0, sd=sd_beta_i)
60
61 proposal_log_joint <- log_joint(curr_beta_0, beta_i_proposal, curr_beta_w, y,
  ↪ intersection, weekend)
62 log_acceptance_prob <- proposal_log_joint - curr_log_joint
63
64 if (log(runif(1)) < log_acceptance_prob) {
65   curr_beta_i <- beta_i_proposal
66   curr_log_joint <- proposal_log_joint
67   acceptance_beta_i <- acceptance_beta_i + 1
68 }
69
70 beta_i_trace[i] <- curr_beta_i
71
72 # kernel 3: propose a new beta_w
73
74 beta_w_proposal <- curr_beta_w + rnorm(n=1, mean=0, sd=sd_beta_w)
75
76 proposal_log_joint <- log_joint(curr_beta_0, curr_beta_i, beta_w_proposal, y,
  ↪ intersection, weekend)
77 log_acceptance_prob <- proposal_log_joint - curr_log_joint
78
79 if (log(runif(1)) < log_acceptance_prob) {
80   curr_beta_w <- beta_w_proposal
81   curr_log_joint <- proposal_log_joint
82   acceptance_beta_w <- acceptance_beta_w + 1
83 }
84
85 beta_w_trace[i] <- curr_beta_w
86 }
87
88 end_time <- Sys.time()
89
90 return(
91   list(
92     beta_0_trace = beta_0_trace,
93     beta_i_trace = beta_i_trace,

```

```

94     beta_w_trace = beta_w_trace,
95     acceptance_beta_0 = acceptance_beta_0 / n_iters,
96     acceptance_beta_i = acceptance_beta_i / n_iters,
97     acceptance_beta_w = acceptance_beta_w / n_iters,
98     runtime = as.numeric(end_time - start_time),
99     last_beta_0 = curr_beta_0,
100    last_beta_i = curr_beta_i,
101    last_beta_w = curr_beta_w
102  )
103 )
104 }
105
106
107 forward <- function(synthetic_data_size, df) {
108   b0 <- rnorm(1, mean = 0, sd = 5)
109   bi <- rnorm(1, mean = 0, sd = 2)
110   bw <- rnorm(1, mean = 0, sd = 1)
111
112   # we will take random samples from the crash dataframe
113
114   df_subset <- df |>
115     slice_sample(n = synthetic_data_size)
116
117   log_odds <- b0 + (bi * df_subset$Intersection.Crash) + (bw * df_subset$weekend)
118
119   y_synthetic <- rbinom(n = synthetic_data_size, size = 1, prob = plogis(log_odds))
120
121   list(
122     y = y_synthetic,
123     intersection = df_subset$Intersection.Crash,
124     weekend = df_subset$weekend,
125     beta_0 = b0,
126     beta_i = bi,
127     beta_w = bw
128   )
129 }
130
131 forward_posterior <- function(synthetic_data_size, n_mcmc_iters, df, param) {
132   initial <- forward(synthetic_data_size, df)
133
134   if (n_mcmc_iters > 0) {
135     samples <- custom_mh(y = initial$y,
136                          intersection = initial$intersection,
137                          weekend = initial$weekend,
138                          n_iters = n_mcmc_iters,
139                          beta_0 = initial$beta_0,
140                          beta_i = initial$beta_i,
141                          beta_w = initial$beta_w)
142
143     list(
144       beta_0 = samples$last_beta_0,
145       beta_i = samples$last_beta_i,

```



```
146     beta_w = samples$last_beta_w
147   )[[param]]
148
149
150 }
151
152 list(
153   beta_0 = initial$beta_0,
154   beta_i = initial$beta_i,
155   beta_w = initial$beta_w
156 )[[param]]
157 }
```

The next file, `modell1_custom_mh.R`, performs 4000 iterations of MCMC (plus 1000 burn-in iterations for a total of 5000) using the custom MH sampler.

```
1  # modell1_custom_mh.R
2  source("modell1_custom_mh_functions.R")
3
4  # load in the crash_weather data
5  df_simple <- read.csv("data/crash_weather.csv")
6
7  set.seed(1234)
8  initial_beta_0 <- rnorm(1, mean = 0, sd = 0.5)
9  initial_beta_i <- rnorm(1, mean = 0, sd = 0.5)
10 initial_beta_w <- rnorm(1, mean = 0, sd = 0.5)
11
12 n <- 5000
13 samples <- custom_mh(y = df_simple$Pedestrian.Flag,
14                      intersection = df_simple$centered_intersection,
15                      weekend = df_simple$weekend,
16                      n_iters = n,
17                      beta_0 = initial_beta_0,
18                      beta_i = initial_beta_i,
19                      beta_w = initial_beta_w)
20
21 beta_0_part_trace <- samples$beta_0_trace[1000:n]
22 beta_i_part_trace <- samples$beta_i_trace[1000:n]
23 beta_w_part_trace <- samples$beta_w_trace[1000:n]
24
25
26 plot(y=beta_0_part_trace, x=1:length(beta_0_part_trace), type="l", col="steelblue",
27      main="Trace Plot for beta_0", xlab="MCMC iteration", ylab="beta_0")
28 plot(y=beta_i_part_trace, x=1:length(beta_i_part_trace), type="l", col="steelblue",
29      main="Trace Plot for beta_i", xlab="MCMC iteration", ylab="beta_i")
30 plot(y=beta_w_part_trace, x=1:length(beta_w_part_trace), type="l", col="steelblue",
31      main="Trace Plot for beta_w", xlab="MCMC iteration", ylab="beta_w")
32
33
34 hist(beta_0_part_trace, main="Posterior histogram for beta_0",
35      xlab="beta_0", col="steelblue", border="white")
36 hist(beta_i_part_trace, main="Posterior histogram for beta_i",
37      xlab="beta_i", col="steelblue", border="white")
38 hist(beta_w_part_trace, main="Posterior histogram for beta_w",
39      xlab="beta_w", col="steelblue", border="white")
40
41
42 mcmc_matrix <- cbind(beta_0_part_trace, beta_i_part_trace, beta_w_part_trace)
43 colnames(mcmc_matrix) <- c("beta_0", "beta_i", "beta_w")
44
45 draws_obj <- as_draws_matrix(mcmc_matrix)
46 fit_summary <- summarise_draws(draws_obj)
47 write.csv(fit_summary, "results/modell1_custom_mh_summary.csv", row.names=F)
```

This next file, `modell1_custom_mh_test.R`, performs the exact invariance test using synthetic datasets and 1000 replications.

```
1  # modell1_custom_mh_test.R
2  source("modell1_custom_mh_functions.R")
3
4  # load in the crash_weather data
5  df <- read.csv("data/crash_weather_full.csv")
6
7
8  set.seed(405)
9  # param = 1 tests beta_0, 2 tests beta_i, 3 tests beta_w
10 n <- 200
11
12 for (param in 1:3) {
13   forward_only <- replicate(1000, forward_posterior(n, 0, df, param))
14
15   with_mcmc <- replicate(1000, forward_posterior(n, 200, df, param))
16
17   t_test <- t.test(forward_only, with_mcmc)
18   print(t_test$p.value)
19
20
21   ks_test <- ks.test(forward_only, with_mcmc)
22   print(ks_test$p.value)
23 }
24
```

This next file, `model1.stan`, provides the Stan code for Model 1, to be used for variational inference.

```
1  // model1.stan
2  data {
3    int<lower=1> N;
4    array[N] int<lower=0, upper=1> y;
5
6    vector[N] intersection;
7    vector[N] weekend;
8  }
9
10
11 parameters {
12   real beta_0; // intercept
13   real beta_i; // intersection effect
14   real beta_w; // weekend effect
15 }
16
17
18 model {
19   beta_0 ~ normal(0, 5);
20   beta_i ~ normal(0, 2);
21   beta_w ~ normal(0, 1);
22
23   y ~ bernoulli_logit(beta_0 + beta_i*intersection + beta_w*weekend);
24 }
```

This file, `modell1_variational.R`, performs variational inference of Model 1 using Stan.

```
1  # modell1_variational.R
2
3  # load in the crash_weather data
4  df_simple <- read.csv("data/crash_weather.csv")
5
6
7  modell1_data <- list(
8    N = nrow(df_simple),
9    y = df_simple$Pedestrian.Flag,
10   intersection = df_simple$centered_intersection,
11   weekend = df_simple$weekend
12 )
13
14
15 modell1 <- cmdstan_model("stan/modell1.stan")
16
17 dir.create(file.path("stan_out"), showWarnings=FALSE)
18
19 fit_variational <- modell1$variational(
20   seed = 405,
21   refresh = 500,
22   output_dir = "stan_out",
23   algorithm = "meanfield",
24   output_samples = 4000,
25   data = modell1_data
26 )
27
28 vi_summary <- fit_variational$summary()
29
30 write.csv(vi_summary, "results/modell1_vi_summary.csv", row.names=F)
31 print(vi_summary)
32
```

This file, `model2.stan`, provides Stan code for model 2, to be used for both HMC and variational inference.

```
1 // model2.stan
2 data {
3   int<lower=1> N;
4   int<lower=1> K;
5   array[N] int<lower=0, upper=1> y;
6
7   matrix[N, K] time_dummies;
8   vector[N] intersection;
9   vector[N] weekend;
10  vector[N] avg_snow_cm;
11  vector[N] avg_rain_mm;
12  vector[N] latitude;
13  vector[N] longitude;
14 }
15
16
17 parameters {
18   real beta_0; // intercept
19   vector[K] beta_t; // time of day effect (there are K+1 time categories)
20   real beta_i; // intersection effect
21   real beta_w; // weekend effect
22   real beta_snow; // snow effect
23   real beta_rain; // rain effect
24   real beta_lat; // latitude effect
25   real beta_lon; // longitude effect
26 }
27
28
29 model {
30   beta_0 ~ normal(0, 5);
31   beta_t ~ normal(0, 1);
32   beta_i ~ normal(0, 2);
33   beta_w ~ normal(0, 1);
34   beta_snow ~ normal(0, 3);
35   beta_rain ~ normal(0, 3);
36   beta_lat ~ normal(0, 2);
37   beta_lon ~ normal(0, 2);
38
39
40   y ~ bernoulli_logit(beta_0 + time_dummies*beta_t +
41                       beta_i*intersection + beta_w*weekend +
42                       beta_snow*avg_snow_cm + beta_rain*avg_rain_mm +
43                       beta_lat*latitude + beta_lon*longitude);
44 }
45
```

Lastly, this file, model2.R, performs HMC and variational inference for Model 2.

```
1 # model2.R
2
3 # load in the crash_weather data
4 df_simple <- read.csv("data/crash_weather.csv")
5
6 timeofday_matrix <- model.matrix(~ timeofday, data=df_simple)[,-1]
7
8 model2_data <- list(
9   N = nrow(df_simple),
10  K = ncol(timeofday_matrix),
11  y = df_simple$Pedestrian.Flag,
12  time_dummies = timeofday_matrix,
13  intersection = df_simple$centered_intersection,
14  weekend = df_simple$weekend,
15  avg_snow_cm = df_simple$avg_snow_cm,
16  avg_rain_mm = df_simple$avg_rain_mm,
17  latitude = df_simple$centered_lat,
18  longitude = df_simple$centered_lon
19 )
20
21 model2 <- cmdstan_model("stan/model2.stan")
22
23 dir.create(file.path("stan_out"), showWarnings=FALSE)
24
25 fit_hmc <- model2$sample(
26   seed = 1,
27   chains = 1,
28   refresh = 500,
29   iter_sampling = 2000,
30   data = model2_data,
31   output_dir = "stan_out"
32 )
33
34 fit_variational <- model2$variational(
35   seed = 1,
36   refresh = 500,
37   output_dir = "stan_out",
38   algorithm = "meanfield",
39   output_samples = 2000,
40   data = model2_data
41 )
42
43 variables <- c("beta_0", "beta_w", "beta_i", "beta_snow", "beta_rain",
44               "beta_t[1]", "beta_t[2]", "beta_t[3]", "beta_t[4]",
45               "beta_lat", "beta_lon")
46
47 hmc_draws <- fit_hmc$draws(variables = variables)
48
49 for (param in variables) {
50   trace_plot <- mcmc_trace(hmc_draws , pars = c(param)) +
51     ggtitle(paste("Trace Plot for", param)) +
```

```

52     theme(plot.title = element_text(hjust = 0.5)) +
53     xlab("MCMC Iteration")
54     print(trace_plot)
55 }
56
57
58 vi_summary <- fit_variational$summary()
59
60 write.csv(vi_summary, "results/model2_vi_summary.csv", row.names=F)
61
62 hmc_summary <- fit_hmc$summary()
63
64 write.csv(hmc_summary, "results/model2_hmc_summary.csv", row.names=F)

```